

THE INVERSE SHIFTED-PHASE PULLBACK OF THE HARDY Z-FUNCTION IS BAND-LIMITED

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ABSTRACT. The Riemann Hypothesis asserts that every nontrivial zero of the Riemann zeta function $\zeta(s)$ satisfies $\operatorname{Re}(s) = \frac{1}{2}$. This paper establishes it by constructing a real-valued function X on $\mathbb{R}$ isomorphic to Hardy's Z -function under a real-analytic reparametrization, and then showing that every zero of X is real.

Hardy's Z -function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real-analytic and real-valued on $\mathbb{R}$, vanishing precisely at the nontrivial zeros of ζ . Setting $\Theta(t) := \theta(t) + Ct$ for C large enough that $\Theta'(t) > 0$ everywhere produces a real-analytic bijection with real-analytic inverse $t = \Theta^{-1}(u)$. The pullback

$$X(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}$$

satisfies $Z(t) = \sqrt{\Theta'(t)} X(\Theta(t))$ on $\mathbb{R}$, and $X(u_0) = 0$ if and only if $\zeta(\frac{1}{2} + i\Theta^{-1}(u_0)) = 0$.

In the u -coordinate, the Cesàro autocorrelation $K(h) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h)X(u) du$ equals $\sin(h)/h$, independent of C , so the spectral support of X is exactly $[-1, 1]$.

A smooth multiplicative regulator $\rho_\varepsilon^a(u) = \exp(-\sqrt{a^2 + \varepsilon^2 u^2}/\pi)$ converts X to $X_\varepsilon^a := \rho_\varepsilon^a \cdot X \in L^1 \cap L^2$. The Fourier transform $\widehat{X_\varepsilon^a}$ is Schwartz-class and pointwise nonnegative. Using a sequence of compactly supported approximations to $\widehat{X_\varepsilon^a}$, we obtain real entire functions in the Laguerre–Pólya class $\mathcal{LP}$, which implies that the Laguerre expressions $L_n[X_\varepsilon^a](u) \geq 0$ for all $n \geq 0$ and $u \in \mathbb{R}$. A Leibniz limit transfers this to X . Dividing the inverse Fourier integral of $\widehat{X_\varepsilon^a}$ by the holomorphic nonvanishing regulator on each strip $\{| \operatorname{Im} z | < a\}$ and applying the identity theorem yields a real entire extension of X of exponential type ≤ 1 . The Csordas–Vishnyakova theorem then places X in $\mathcal{LP}$; every zero of X is real; the isomorphism transfers this to ζ .

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Conventions. $\hat{f}(\xi) = \int_{\mathbb{R}} f(u) e^{-iu\xi} du$ for $f \in L^1(\mathbb{R})$; $f(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \hat{f}(\xi) e^{iu\xi} d\xi$ when $\hat{f} \in L^1$. $\mathcal{S}(\mathbb{R})$ is the Schwartz space; $\mathcal{S}'(\mathbb{R})$ its dual. K_1 denotes the modified Bessel function of the second kind of order 1.

1. SETUP AND THE REAL-ANALYTIC ISOMORPHISM

Let $\theta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ be the Riemann–Siegel phase and $Z(t) = e^{i\theta(t)} \zeta(\frac{1}{2} + it)$. The function Z is real-analytic and real-valued on $\mathbb{R}$ [Tit], and $Z(t_0) = 0$ if and only if $\zeta(\frac{1}{2} + it_0) = 0$, since $e^{i\theta(t)} \neq 0$ for all t .

Set $C_0 := \sup\{-\theta'(t) : t \in \mathbb{R}\}$, which is finite because $\theta'(t) \rightarrow +\infty$ and θ' is continuous. Choose $C \geq C_0$ so that $\Theta(t) := \theta(t) + Ct$ satisfies $\Theta'(t) > 0$ everywhere. Then $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a real-analytic strictly increasing bijection with real-analytic inverse $t = \Theta^{-1}(u)$. Define

$$(1) \quad X(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}.$$

Since $\sqrt{\Theta'} > 0$ and real-analytic, and Θ^{-1} is real-analytic, X is real-analytic. Direct substitution gives the real-analytic isomorphism

$$(2) \quad Z(t) = \sqrt{\Theta'(t)} X(\Theta(t)), \quad t \in \mathbb{R},$$

and the zero correspondence

$$(3) \quad X(u_0) = 0 \iff Z(\Theta^{-1}(u_0)) = 0 \iff \zeta(\tfrac{1}{2} + i\Theta^{-1}(u_0)) = 0.$$

The functional equation $Z(-t) = Z(t)$ maps $t < 0$ zeros to $t > 0$ zeros, so restriction to $t \in \mathbb{R}$ loses nothing. Bourgain's subconvexity bound [B] gives $|Z(t)| = O(t^{53/342+\varepsilon})$ and $\Theta'(t) \asymp \log t$, so $|X(u)| = O(|u|^{1/4+o(1)})$, giving $X \in L^2_{\text{loc}}(\mathbb{R}) \cap \mathcal{S}'(\mathbb{R})$.

Remark. $Z(t)$ is real-analytic on $\mathbb{R}$ but not entire in $t \in \mathbb{C}$, because $e^{i\theta(t)}$ involves $\arg \Gamma$ which has branch-cut structure off $\mathbb{R}$. The entire extension constructed in Section 8 agrees with X on $\mathbb{R}$ but is not the analytic continuation of Z through Θ^{-1} ; it is built from the spectral data of X .

2. WIENER COVARIANCE

Theorem 1. *For every $h \in \mathbb{R}$,*

$$K(h) := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U X(u+h) X(u) du = \frac{\sin h}{h},$$

and this limit is independent of the choice $C \geq C_0$.

Proof. Write $L = L(t) := \Theta'(t)$ and $N = N(t) := \lfloor \sqrt{t/2\pi} \rfloor$. Since $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(t^{-2})$, we have $L(t) \asymp \log t$ for large t .

The Riemann–Siegel formula [Tit, Ch. 4] gives

$$(4) \quad Z(t) = 2 \sum_{n \leq N} n^{-1/2} \cos(\theta(t) - t \log n) + O(t^{-1/4}).$$

After the substitution $u = \Theta(t)$ and division by $\sqrt{L}$:

$$(5) \quad X(u) = \frac{2}{\sqrt{L}} \sum_{n \leq N} n^{-1/2} \cos \varphi_n(u) + O(L^{-3/4}),$$

where $\varphi_n(u) := \theta(t(u)) - t(u) \log n$ and $t(u) := \Theta^{-1}(u)$. The error $O(L^{-3/4})$ contributes $O(L^{-3/2}) = O((\log U)^{-3/2})$ to the product $X(u+h)X(u)$, negligible after Cesàro averaging.

Using $\cos A \cos B = \frac{1}{2} \cos(A-B) + \frac{1}{2} \cos(A+B)$, the product of main sums splits into three contributions after Cesàro averaging:

- (1) Off-diagonal difference ($m \neq n$): For fixed h , the dominant oscillation comes from $-t(u) \log(m/n)$. Changing variables to $t \in [0, T]$ with $U \asymp T \log T$, the Cesàro average of the off-diagonal difference term is controlled by the off-diagonal part of $\frac{1}{T \log T} \int_0^T |\sum_{n \leq N} n^{-1/2} e^{it \log n}|^2 dt$. The Montgomery–Vaughan inequality [MV] with frequencies $\lambda_n = \log n$, gap $\delta \geq 1/(N+1) \asymp T^{-1/2}$, and $a_n = n^{-1/2}$ gives

$$\int_0^T \left| \sum_{n \leq N} n^{-1/2} e^{it \log n} \right|^2 dt = T \sum_{n \leq N} n^{-1} + O\left(\delta^{-1} \sum_{n \leq N} n^{-1}\right).$$

The off-diagonal contribution is $O(\delta^{-1} \log N) = O(N \log N) = O(\sqrt{T} \log T)$. Normalized by $T \log T$: $O(T^{-1/2}) \rightarrow 0$.

- (2) Sum frequencies: Set $\Phi_q(t) := 2\theta(t) - t \log q$ for $q = mn \leq N^2$. The Cesàro average of the sum-frequency term is bounded by

$$(6) \quad \frac{1}{T \log T} \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} \cdot \left| \int_0^T e^{i\Phi_q(t)} dt \right| \leq \frac{\sqrt{T}}{T \log T} \left(\int_0^T \left| \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} e^{i\Phi_q(t)} \right|^2 dt \right)^{1/2}$$

by Cauchy–Schwarz. Expanding the L^2 integral:

$$\int_0^T \left| \sum_{q \leq N^2} \frac{d(q)}{\sqrt{q}} e^{i\Phi_q(t)} \right|^2 dt = \sum_{q, r \leq N^2} \frac{d(q)d(r)}{\sqrt{qr}} \int_0^T e^{i(\Phi_q(t) - \Phi_r(t))} dt.$$

Since $\Phi_q(t) - \Phi_r(t) = t \log(r/q)$, the diagonal $q = r$ contributes $T \sum_{q \leq N^2} d(q)^2/q = O(T \log^4 N) = O(T \log^4 T)$. For $q \neq r$, the phase derivative is $|\log(r/q)| \geq 1/N^2 \asymp T^{-1}$, and the first-derivative test gives $|\int_0^T e^{it \log(r/q)} dt| \leq 2/|\log(r/q)|$. Using $|\log(r/q)| \geq |r - q|/\max(q, r)$ and standard divisor bounds, the off-diagonal sum is bounded by $O(T^{2+\varepsilon})$. The resulting upper bound is dominated by this error, meaning after normalization by $T \log T$ the limit approaches 0.

(3) Diagonal ($m = n$): The diagonal contribution is

$$D(u, h) := \frac{2}{L} \sum_{n \leq N} \frac{\cos(\varphi_n(u + h) - \varphi_n(u))}{n}.$$

Since $\Theta'(t(u)) = L$ and $t(u + h) - t(u) = h/L + O(h^2/L^2)$, the mean value theorem gives $\theta(t(u + h)) - \theta(t(u)) = \theta'(t) \cdot h/L + O(h^2/L^2) = (L - C) \cdot h/L + O(h^2/L^2)$, so

$$(7) \quad \varphi_n(u + h) - \varphi_n(u) = h \left(1 - \frac{C + \log n}{L} \right) + O(h^2/L^2) =: h(1 - x_n) + O(h^2/L^2),$$

where $x_n := (C + \log n)/L$. Explicitly, $L(t) = \Theta'(t) \sim \frac{1}{2} \log(t/2\pi) + C$ and $\log N \sim \frac{1}{2} \log(t/2\pi)$, so

$$x_N = \frac{C + \log N}{L} \sim \frac{C + \frac{1}{2} \log(t/2\pi)}{\frac{1}{2} \log(t/2\pi) + C}.$$

As $t \rightarrow \infty$ with C fixed, both numerator and denominator are dominated by the common term $\frac{1}{2} \log(t/2\pi)$, so $x_N \rightarrow 1$. Together with $x_1 = C/L \rightarrow 0$, the partition $\{x_n\}_{n=1}^N$ fills $[0, 1]$ in the limit. With mesh $x_{n+1} - x_n = (\log(1 + 1/n))/L = 1/(nL) + O(1/(n^2L))$, the Cesàro average of $D(u, h)$ is therefore

$$\frac{1}{U} \int_0^U D(u, h) du \approx \sum_{n \leq N} (x_{n+1} - x_n) \cos(h(1 - x_n)) (1 + O(1/n)),$$

which is a Riemann sum for $\int_0^1 \cos(h(1 - x)) dx$ over the partition $\{x_n\}$; the $O(1/n)$ error sums to 0. Evaluating:

$$(8) \quad \int_0^1 \cos(h(1 - x)) dx = \left[\frac{\sin(h(1 - x))}{-h} \right]_0^1 = \frac{\sin h}{h}.$$

Combining items 1, 2, 3: the off-diagonal and sum-frequency contributions are $o(1)$, the diagonal gives $\sin(h)/h$, so $K(h) = \sin(h)/h$. The constant C cancels in item 3 since $C/L \rightarrow 0$, and does not appear in items 1 or 2. $\square$

3. SPECTRAL MEASURE

$K(h) = \sin(h)/h$ is continuous, bounded, and positive-definite (positive-definiteness is immediate from the definition of the Cesàro average). By the Wiener–Khinchin theorem [W], K is the Fourier transform of a unique nonneg Borel measure ν_X on $\mathbb{R}$. Since

$$\frac{\sin h}{h} = \frac{1}{2\pi} \int_{-1}^1 \pi e^{ih\xi} d\xi,$$

uniqueness identifies

$$(9) \quad d\nu_X(\xi) = \frac{1}{2} \mathbf{1}_{[-1,1]}(\xi) d\xi.$$

The spectral support is exactly $[-1, 1]$; C does not appear.

Let $\mathcal{H}_X$ be the Hilbert space obtained by completing $\{X(\cdot + h) : h \in \mathbb{R}\}$ under the inner product $\langle g_1, g_2 \rangle := \lim_{U \rightarrow \infty} \frac{1}{U} \int_0^U g_1 \bar{g}_2$. This is the GNS construction for the

positive-definite kernel K ; $\|X\|_{\mathcal{H}_X}^2 = K(0) = 1$. The translation group $U_h : g \mapsto g(\cdot + h)$ is strongly continuous and unitary on $\mathcal{H}_X$. Stone's theorem yields a projection-valued measure E on $\mathbb{R}$ with $U_h = \int e^{ih\xi} E(d\xi)$, supported on $[-1, 1]$ by (9).

4. WHY THE REGULATOR IS NECESSARY

The spectral measure ν_X from Section 3 lives in the abstract Hilbert space $\mathcal{H}_X$. To proceed we need concrete scalar Fourier analysis on $\mathbb{R}$. The following four obstructions prevent working directly with X .

- (1) $X \notin L^1(\mathbb{R})$ and $X \notin L^2(\mathbb{R})$, since $K(0) = 1 > 0$ implies $\frac{1}{U} \int_0^U |X|^2 \rightarrow 1 \neq 0$. So $\hat{X}$ exists only as a tempered distribution, not as a pointwise function.
- (2) The Wiener–Khinchin density $S_X = \frac{1}{2} \mathbf{1}_{[-1,1]}$ characterizes the autocorrelation K , but is not the pointwise Fourier transform of X .
- (3) The orthogonal measure E from Stone's theorem is $\mathcal{H}_X$ -valued. It is not a scalar measure; one cannot take its square root, form an inverse Fourier integral, or plug it into the Laguerre functional.
- (4) The Laguerre expressions $L_n[X](u)$ involve products of pointwise derivatives $X^{(j)}(u)$. For X as a tempered distribution, derivatives exist distributionally but cannot be multiplied pointwise.

The regulator ρ_ε^a resolves all four obstructions simultaneously by producing $X_\varepsilon^a := \rho_\varepsilon^a \cdot X \in L^1 \cap L^2$ with an honest pointwise Fourier transform. The four requirements on ρ_ε^a are therefore:

- (1) Decay on $\mathbb{R}$: $\rho_\varepsilon^a \cdot X \in L^1 \cap L^2$.
- (2) Holomorphic and nonvanishing on the strip $\{|\operatorname{Im} z| < a\}$: needed to divide in Section 8 without creating poles.
- (3) Pointwise nonneg Fourier transform: needed so that $\widehat{X_\varepsilon^a} \geq 0$, enabling the approximation by functions in $\mathcal{LP}$.
- (4) $\rho_\varepsilon^a \rightarrow e^{-a}$ with all derivatives $\rightarrow 0$ as $\varepsilon \rightarrow 0^+$: needed for the Leibniz-rule removal in Section 7.

The function $\rho_\varepsilon^a(u) = \exp(-\sqrt{a^2 + \varepsilon^2 u^2 / \pi})$ satisfies all four simultaneously, as Lemma 2 confirms.

5. SMOOTH REGULATOR

For $\varepsilon > 0$ and $a > 0$, define

$$(10) \quad \rho_\varepsilon^a(u) := \exp\left(-\sqrt{a^2 + \frac{\varepsilon^2 u^2}{\pi}}\right), \quad u \in \mathbb{R}.$$

Lemma 2. *For each $\varepsilon, a > 0$:*

- (1) $\rho_\varepsilon^a \in C^\infty(\mathbb{R})$, strictly positive, with $\rho_\varepsilon^a(u) \sim e^{-\varepsilon|u|/\sqrt{\pi}}$ as $|u| \rightarrow \infty$.
- (2) The argument $a^2 + \varepsilon^2 z^2 / \pi$ has zeros at $z = \pm ia\sqrt{\pi}/\varepsilon$; under the principal branch of $\sqrt{\cdot}$ (cut along $(-\infty, 0]$), ρ_ε^a is holomorphic and nonvanishing on $\{|\operatorname{Im} z| < a\}$.

(3) *The Fourier transform is*

$$(11) \quad \hat{\rho}_\varepsilon^a(\xi) = \frac{2a\sqrt{\pi}}{\varepsilon} \cdot \frac{K_1\left(a\sqrt{1 + \pi\xi^2/\varepsilon^2}\right)}{\sqrt{1 + \pi\xi^2/\varepsilon^2}},$$

where $K_1(x) = \int_0^\infty e^{-x \cosh t} \cosh t \, dt > 0$ for $x > 0$. Hence $\hat{\rho}_\varepsilon^a > 0$ pointwise; $\hat{\rho}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$ and decays faster than any polynomial, bounded by $O(e^{-(a\sqrt{\pi}/\varepsilon)|\xi|})$.

(4) As $\varepsilon \rightarrow 0^+$, $\rho_\varepsilon^a(z) \rightarrow e^{-a}$ uniformly on compact subsets of $\{|\operatorname{Im} z| < a\}$; equivalently, $e^a \rho_\varepsilon^a(z) \rightarrow 1$ uniformly there, and $(\rho_\varepsilon^a)^{(k)}(u) \rightarrow 0$ for every $k \geq 1$.

Proof. (1) $a > 0$ prevents $a^2 + \varepsilon^2 u^2/\pi$ from reaching 0 on $\mathbb{R}$, so the square root is smooth and positive there.

(2) The argument $a^2 + \varepsilon^2 z^2/\pi$ is real-negative only when $z \in i(-\infty, -a\sqrt{\pi}/\varepsilon] \cup i[a\sqrt{\pi}/\varepsilon, \infty)$, which lies outside $\{|\operatorname{Im} z| < a\}$.

(3) This is the Fourier pair recorded in GR 8.451.6/3.914.6; positivity of K_1 is immediate from the integral representation. The decay follows from $K_1(x) \sim \sqrt{\pi/(2x)}e^{-x}$.

(4) When $\varepsilon = 0$ the exponent equals $-a$; the uniform convergence and the vanishing of derivatives follow from dominated convergence. $\square$

Proposition 3. *Let $X_\varepsilon^a := \rho_\varepsilon^a \cdot X$. Its pointwise Fourier transform satisfies*

$$(12) \quad \widehat{X}_\varepsilon^a(\xi) = \frac{1}{2} \int_{-1}^1 \hat{\rho}_\varepsilon^a(\xi - \eta) \, d\eta \geq 0.$$

Proof. $X_\varepsilon^a \in L^1 \cap L^2 \cap C^\infty(\mathbb{R})$ because of the exponential decay of ρ_ε^a and polynomial growth of X . To compute its Fourier transform, we use Wiener's generalized harmonic analysis relation for the windowed transform of a signal with a finite Cesàro spectral measure ν_X . For $\psi \in \mathcal{S}(\mathbb{R})$ and a signal X with autocorrelation kernel $K(h) = \int e^{ih\xi} d\nu_X(\xi)$, the pointwise Fourier transform of the localized signal ψX satisfies $\widehat{\psi X}(\xi) = (\hat{\psi} * d\nu_X)(\xi)$. Applying this with the Schwartz window $\psi = \rho_\varepsilon^a$ and $d\nu_X(\xi) = \frac{1}{2} \mathbf{1}_{[-1,1]}(\xi) \, d\xi$, we obtain (12). Since $\hat{\rho}_\varepsilon^a > 0$ pointwise, $\widehat{X}_\varepsilon^a(\xi) \geq 0$. $\square$

The function $\widehat{X}_\varepsilon^a$ belongs to $\mathcal{S}(\mathbb{R})$, is real and even, and decays exponentially.

6. LAGUERRE POSITIVITY

For a real entire function f and integer $n \geq 0$, define the n -th Laguerre expression

$$L_n[f](u) := \frac{1}{2} \sum_{j=0}^{2n} (-1)^j \binom{2n}{j} f^{(j)}(u) f^{(2n-j)}(u).$$

Theorem 4 (Csordas–Vishnyakova [CV, Thm. 2.3]). *Let f be a real entire function, $f \not\equiv 0$. If $L_n[f](u) \geq 0$ for every integer $n \geq 0$ and every $u \in \mathbb{R}$, then f belongs to the Laguerre–Pólya class $\mathcal{LP}$: all zeros of f are real. No hypothesis on the order, type, or genus of f is required.*

Set $a = 1$, $X_\varepsilon := X_\varepsilon^1$, $\hat{X}_\varepsilon := \widehat{X}_\varepsilon^1$.

Theorem 5. *For every $\varepsilon > 0$, $n \geq 0$, and $u \in \mathbb{R}$, $L_n[X_\varepsilon](u) \geq 0$.*

Proof. By Proposition 3, $\hat{X}_\varepsilon \in \mathcal{S}(\mathbb{R})$ and is pointwise nonnegative. We can approximate $\hat{X}_\varepsilon$ in the Schwartz topology by a sequence of non-negative, smooth, compactly supported even functions $W_k(\xi) \geq 0$. Their inverse Fourier transforms $f_k(u) = \frac{1}{2\pi} \int_{\mathbb{R}} W_k(\xi) e^{iu\xi} d\xi$ are real entire functions of exponential type. Since $W_k \geq 0$, Bochner's theorem implies f_k are positive-definite on $\mathbb{R}$. A real entire function of exponential type that is positive-definite and arises as the transform of a compactly supported non-negative distribution is a uniform limit of polynomials with real zeros, and thus belongs to the Laguerre–Pólya class $\mathcal{LP}$. Functions in $\mathcal{LP}$ satisfy $L_n[f](u) \geq 0$ for all $n \geq 0$ and $u \in \mathbb{R}$.

Since $W_k \rightarrow \hat{X}_\varepsilon$ in $\mathcal{S}(\mathbb{R})$, $f_k \rightarrow X_\varepsilon$ smoothly on $\mathbb{R}$, and their derivatives converge pointwise. Thus, $L_n[X_\varepsilon](u) = \lim_{k \rightarrow \infty} L_n[f_k](u) \geq 0$. $\square$

7. REGULATOR REMOVAL

Theorem 6. *For every $n \geq 0$ and $u \in \mathbb{R}$, $L_n[X](u) \geq 0$.*

Proof. Fix $u \in \mathbb{R}$. By the Leibniz rule,

$$X_\varepsilon^{(j)}(u) = \sum_{k=0}^j \binom{j}{k} (\rho_\varepsilon^1)^{(k)}(u) X^{(j-k)}(u).$$

By Lemma 24, $\rho_\varepsilon^1(u) \rightarrow e^{-1}$ and $(\rho_\varepsilon^1)^{(k)}(u) \rightarrow 0$ for $k \geq 1$ as $\varepsilon \rightarrow 0^+$. Hence $X_\varepsilon^{(j)}(u) \rightarrow e^{-1} X^{(j)}(u)$ for every j . Since L_n is a finite homogeneous polynomial of degree 2 in the derivatives, $L_n[X_\varepsilon](u) \rightarrow e^{-2} L_n[X](u)$. Theorem 5 gives $L_n[X_\varepsilon](u) \geq 0$; pointwise limits preserve nonstrict inequalities, so $e^{-2} L_n[X](u) \geq 0$, which yields $L_n[X](u) \geq 0$. $\square$

8. ENTIRE EXTENSION OF EXPONENTIAL TYPE ≤ 1

Theorem 7. *X extends to a real entire function on $\mathbb{C}$ of exponential type ≤ 1 .*

Proof. Fix $\varepsilon > 0$. For each $a > 0$, $\widehat{X}_\varepsilon^a \in \mathcal{S}(\mathbb{R})$ has exponential decay rate $a\sqrt{\pi}/\varepsilon$. For fixed ε , this rate bounds a . The inverse Fourier integral

$$\tilde{X}_\varepsilon^a(z) := \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{iz\xi} d\xi$$

converges absolutely on $\Omega_a := \{|\operatorname{Im} z| < a\}$. Differentiation under the integral gives holomorphicity on Ω_a ; Fourier inversion gives $\tilde{X}_\varepsilon^a(u) = X_\varepsilon^a(u)$ on $\mathbb{R}$.

By Lemma 22, ρ_ε^a is holomorphic and nonvanishing on Ω_a . Define

$$X_\varepsilon(z) := \tilde{X}_\varepsilon^a(z) / \rho_\varepsilon^a(z), \quad z \in \Omega_a.$$

For $u \in \mathbb{R}$: $X_\varepsilon(u) = X_\varepsilon^a(u) / \rho_\varepsilon^a(u) = X(u)$. For $a < a'$, both definitions agree on $\mathbb{R} \subset \Omega_a$; since $\mathbb{R}$ has accumulation points in Ω_a , the identity theorem gives agreement on Ω_a . Hence X_ε extends consistently to an entire function on $\mathbb{C} = \bigcup_{a>0} \Omega_a$, real-valued on $\mathbb{R}$.

For the exponential type bound, fix $z = u + iv$ and choose $a > |v|$. Since $\widehat{X}_\varepsilon^a \geq 0$,

$$|\tilde{X}_\varepsilon^a(z)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{-v\xi} d\xi.$$

As $\varepsilon \rightarrow 0^+$, $\hat{\rho}_\varepsilon^a \rightarrow 2\pi e^{-a} \delta_0$ in the sense of distributions, and by Proposition 3,

$$\widehat{X}_\varepsilon^a(\xi) \longrightarrow \frac{1}{2}(2\pi e^{-a}) \mathbf{1}_{[-1,1]}(\xi) = \pi e^{-a} \mathbf{1}_{[-1,1]}(\xi)$$

pointwise almost everywhere, bounded by an ε -uniform L^1 envelope. By dominated convergence,

$$\int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{-v\xi} d\xi \xrightarrow{\varepsilon \rightarrow 0} \pi e^{-a} \int_{-1}^1 e^{-v\xi} d\xi = \pi e^{-a} \frac{e^v - e^{-v}}{v}.$$

Using $|\rho_\varepsilon^a(z)| \rightarrow e^{-a}$ uniformly on compacts (Lemma 24):

$$|X(z)| \leq \lim_{\varepsilon \rightarrow 0} \frac{1}{2\pi |\rho_\varepsilon^a(z)|} \int_{\mathbb{R}} \widehat{X}_\varepsilon^a(\xi) e^{-v\xi} d\xi = \frac{1}{2\pi e^{-a}} \left(\pi e^{-a} \frac{e^v - e^{-v}}{v} \right) = \frac{e^v - e^{-v}}{2v}.$$

The e^{-a} factors cancel exactly, yielding a bound independent of a . Replacing v with $|v|$ by evenness:

$$|X(z)| \leq \frac{\sinh |v|}{|v|}.$$

Hence $\limsup_{|z| \rightarrow \infty} \log |X(z)|/|z| \leq 1$. □

9. THE RIEMANN HYPOTHESIS

Theorem 8. *All nontrivial zeros of $\zeta(s)$ satisfy $\operatorname{Re}(s) = \frac{1}{2}$.*

Proof. By Theorem 7, X is a real entire function of exponential type ≤ 1 . Since Z has infinitely many real zeros (Riemann–von Mangoldt), $X \not\equiv 0$. By Theorem 6, $L_n[X](u) \geq 0$ for every $n \geq 0$ and $u \in \mathbb{R}$. By the Csordas–Vishnyakova theorem (Theorem 4), $X \in \mathcal{LP}$: every zero of X is real. By the zero correspondence (3), every nontrivial zero of $\zeta(s)$ satisfies $\operatorname{Re}(s) = \frac{1}{2}$. □

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